SMART MEDICINE DISPENSER

A portable medication-management concept paired with the DailyMed companion app.

Our concept

A portable smart medicine dispenser that reminds users to take medicines on time, automatically opens the designated compartment, and provides alerts to support a more organized medication routine.

Project stage: Student prototype in development

Designed to explore how simple technology can support everyday medication routines.

PROJECT DETAILS

Location
Imphal, Manipur, India

Email
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Mobile
9233171544

Website
Not available yet — project in development

DAILYMED

Companion app concept for medication schedules and notifications.

OUR TEAM

Five members, distinct contributions

Each team member contributes a specific skill to developing and communicating the project.

Taanish Tourangbam

Role: PDF preparation

Why suitable: Contributes to organizing the project information into a clear, structured document that communicates the team's work professionally.

Soubam Leishem

Role: Hardware programming

Why suitable: Contributes programming skills to coordinate scheduled operations and the behaviour of the device's electronic components.

Moirnagthem Chingkhei

Role: Electrical wiring and hardware connections

Why suitable: Contributes practical electronics skills by connecting components and helping integrate the hardware into a functioning prototype.

Kamini Kangjam

Role: Product design

Why suitable: Contributes design thinking to the product's appearance, compartment layout and portability, helping make the prototype practical and easy to understand.

Lingjelthoibi Ningthoujam

Role: Concept and coordination

Why suitable: Contributes by presenting the ideas, coordinating product features and helping develop the concept into a clear, purposeful project.

Team structure

Five-member student founding team. No equity has been allocated. Each member contributes through their assigned role.

THE NEED

Making medication management simpler

The challenge

- Maintaining a regular medication schedule can be difficult for people managing several prescribed medicines, older adults and people who need additional support.
- Relying only on memory or external reminders can make it harder to keep track of scheduled doses.

What people may use today

- Memory, phone alarms or clocks.
- Ordinary pillboxes and manually organized compartments.
- Reminders from family members or caregivers.

Our proposed solution

The Smart Medicine Dispenser combines timed reminders, designated medicine compartments, automatic compartment access, alerts and the DailyMed app concept in one portable system.

Why this approach may help

- Brings medicine storage and scheduled reminders together.
- Helps users identify the compartment intended for a scheduled dose.
- Provides audio and visual alerts.
- Explores a connected experience between a physical device and an app.
- Aims to balance storage capacity with portability.

Objective: Explore how accessible technology can make medication routines more organized, convenient and easier to manage. The device supports routines; it does not verify that medicine has been swallowed or replace professional medical guidance.

WHAT MAKES IT DISTINCT

A connected concept with room to grow

Integrated medication management

Combines medicine storage and scheduled reminders instead of relying on separate tools alone.

Automatic compartment access

Uses a programmed mechanism to open the designated compartment at the scheduled time.

Multiple alert methods

Combines audible alerts, indicator lights and a display to communicate device status.

Sensor-supported monitoring

Uses sensors to detect compartment access and support the programmed alert process.

Portable physical design

A medium-sized form aims to balance useful storage space with ease of carrying.

DailyMed app concept

Adds a digital interface for medication schedules and notifications, with exact app functions to be validated.

Innovation, honestly stated

What part is novel?

The novelty we are exploring is the integration of existing electronic components and programmed functions into a coordinated, portable medication-management system with a planned companion app. We do not claim that every individual component or feature is newly invented.

Potential as a platform technology

The reminder, scheduled-access and sensor-monitoring concept could potentially be adapted for other organized dispensing or storage applications. Such uses would require separate development and evaluation.

Our advantage

We aim to demonstrate a practical combination of features in a portable prototype, emphasizing clear alerts, usability and the opportunity for app integration. We recognize that existing products already offer some similar functions.

THE PRODUCT

Smart Medicine Dispenser + DailyMed

Physical device features

- **Three separate compartments** for the planned medicine schedule.
- **Timed reminders** when a programmed medication time arrives.
- **Automatic opening and closing** of the designated compartment.
- **Compartment sensors** to detect access.
- **Display, LEDs and buzzer** for visual and audio alerts.
- **Temperature and humidity readings** for environmental awareness.
- **Rechargeable power arrangement** and medium-sized portable design.
- **Optional SOS feature** to signal a need for assistance.

DailyMed app concept

- **Medicine schedule:** add medicine name and time.
- **Time-of-day reminders:** morning, afternoon, evening and night.
- **Notifications:** medicine-time and missed-dose alerts.
- **Optional device alerts:** low battery or disconnected device.
- **Caregiver alerts:** send a missed-dose alert to a chosen caregiver.
- **Possible remote schedule view:** for a caregiver, if implemented.

Working process: Set the medication schedule → receive a reminder → the designated compartment opens → sensors monitor compartment access → the device and app provide configured alerts.

App features beyond the confirmed notes should be treated as planned until implemented. Compartment access alone cannot confirm that a dose has been swallowed.

FUTURE SUSTAINABILITY

A possible path from innovation to impact

Our immediate purpose is to develop and present our innovation. We are not currently operating a business, but the concept could have future practical applications.

1. Individual use

A future device for individuals and families who want additional support with medication routines.

2. Institutional use

Possible evaluation by care homes or other appropriate care settings after further testing.

3. Maintenance and improvements

Repairs, replacement components and future improvements if the product is developed commercially.

4. Connected product development

Explore how DailyMed and future connected features could add value to the physical dispenser.

Who might pay, and to whom?

If the project eventually becomes a business, individuals, families or suitable institutions could purchase the device from its developer or an authorized seller. Future service fees would depend on the final product and service arrangements.

Current status: No prices, sales or revenue have been established. Our present priority is innovation, prototyping, learning and demonstration.

TARGET USERS

Designed around a real-world need

Potential users

- Older adults who want additional support with medication routines.
- Individuals managing multiple prescribed medicines.
- Families and caregivers who help others organize medication schedules.
- Care homes and assisted-living facilities that may wish to evaluate medication-management aids.

Where we could begin

Imphal and other parts of Manipur would be a practical starting point for demonstrating the prototype and gathering feedback. Wider opportunities across India could be explored in the future.

Market opportunity

The project addresses a practical need: helping people organize scheduled medication routines. Potential audiences include individual users, families and care organizations.

At this stage, we have not conducted formal market research or calculated market size. We would need to investigate existing products, user needs, affordability and the costs of producing a reliable device before estimating the opportunity.

Our next step

Demonstrate the prototype, gather feedback, identify improvements and evaluate whether the device would be useful in real-world settings.

Our immediate goal

We want to demonstrate our ideas, technical abilities and teamwork through a student innovation project. Commercial development is a possible future direction, not our current commitment.

LOOKING AHEAD

A three-year development roadmap

An aspirational plan, subject to our team's time, resources and learning.

YEAR 1

Complete and demonstrate

- Develop and demonstrate the prototype's core functions.
- Test timing, compartment mechanisms, alerts and sensors.
- Present the project at innovation events and document our learning.
- Collect feedback and identify improvements.

YEAR 2

Refine the design

- Improve portability, usability and reliability.
- Develop and evaluate the DailyMed app concept.
- Address limitations found during prototype testing.
- Explore feedback from teachers, potential users and relevant professionals.

YEAR 3

Explore future possibilities

- Evaluate feasibility of a more mature prototype.
- Explore suitable educational or care-setting demonstrations.
- Assess manufacturing costs, safety and applicable requirements.
- Decide whether to continue as an educational project or explore further development.

Our vision

To create a thoughtful, portable medication-management aid that reflects our creativity, technical learning and commitment to solving practical problems through innovation.